Wall angle: estimated angle and computed angle

2026-10-24 x–z plane ($\theta_x = 0$) first, three axes at the end of each part

Part 1 Estimated angle

1 Measurement coordinate and local coordinate

$$V = R_y(\theta_y)\, R_x(\theta_x) = [\, \mathbf{u} \;\; \mathbf{v} \;\; \mathbf{w} \,]$$

$$\theta_x = 0: \quad \delta\bar{x} = \cos\theta_y\, \delta\bar{u} + \sin\theta_y\, \delta\bar{w}$$

$$\delta\bar{z} = -\sin\theta_y\, \delta\bar{u} + \cos\theta_y\, \delta\bar{w}$$

2 Random component of the measured tracking error

$$\overline{\delta\bar{\mathbf{p}}_m}[k+1] = a_{\mathrm{pd}}\, \delta\bar{\mathbf{p}}_m[k] + (1 - a_{\mathrm{pd}})\, \overline{\delta\bar{\mathbf{p}}_m}[k]$$

$$\delta\bar{\mathbf{p}}_r[k] = \delta\bar{\mathbf{p}}_m[k] - \overline{\delta\bar{\mathbf{p}}_m}[k] = [\, \delta\bar{x}_r \;\; \delta\bar{y}_r \;\; \delta\bar{z}_r \,]^{\mathsf{T}}$$

3 Variance and covariance in the local coordinate

assumption: the random components along the wall and along the surface normal are uncorrelated

$$\mathrm{E}\big[\delta\bar{u}_r^2\big] = C_T\, \kappa_T\, \bar{a}_T + C_n\, \sigma_n^2$$

$$\mathrm{E}\big[\delta\bar{w}_r^2\big] = C_T\, \kappa_T\, \bar{a}_N + C_n\, \sigma_n^2$$

$$\mathrm{E}\big[\delta\bar{u}_r\, \delta\bar{w}_r\big] = 0$$

4 Estimated local coordinate

the estimator uses $\hat{\theta}_y$; the wall is at θ_y

$$e_\theta = \theta_y - \hat{\theta}_y$$

$$[\, \delta\tilde{u}_r \;\; \delta\tilde{w}_r \,]^{\mathsf{T}} = R_y(\hat{\theta}_y)^{\mathsf{T}}\, [\, \delta\bar{x}_r \;\; \delta\bar{z}_r \,]^{\mathsf{T}}$$

$$= R_y(\hat{\theta}_y)^{\mathsf{T}}\, R_y(\theta_y)\, [\, \delta\bar{u}_r \;\; \delta\bar{w}_r \,]^{\mathsf{T}} = R_y(e_\theta)\, [\, \delta\bar{u}_r \;\; \delta\bar{w}_r \,]^{\mathsf{T}}$$

$$\delta\tilde{u}_r = \cos e_\theta\, \delta\bar{u}_r + \sin e_\theta\, \delta\bar{w}_r$$

$$\delta\tilde{w}_r = -\sin e_\theta\, \delta\bar{u}_r + \cos e_\theta\, \delta\bar{w}_r$$

5 Covariance in the estimated local coordinate and the measured angle

$$\delta\tilde{u}_r\,\delta\tilde{w}_r = -\sin e_\theta \cos e_\theta\,\delta\bar{u}_r^2 + (\cos^2 e_\theta - \sin^2 e_\theta)\,\delta\bar{u}_r\,\delta\bar{w}_r + \sin e_\theta \cos e_\theta\,\delta\bar{w}_r^2$$

$$\mathrm{E}\big[\delta\tilde{u}_r\,\delta\tilde{w}_r\big] = -C_T\,\kappa_T\,(\bar{a}_T - \bar{a}_N)\,\sin e_\theta \cos e_\theta$$

$\sin e_\theta \approx e_\theta$, $\cos e_\theta \approx 1$

$$\mathrm{E}\big[\delta\tilde{u}_r\,\delta\tilde{w}_r\big] \approx -C_T\,\kappa_T\,(\bar{a}_T - \bar{a}_N)\,e_\theta$$

$$e_\theta \approx -\frac{\mathrm{E}\big[\delta\tilde{u}_r\,\delta\tilde{w}_r\big]}{C_T\,\kappa_T\,(\bar{a}_T - \bar{a}_N)}$$

one sample in place of $\mathrm{E}[\,\cdot\,]$; estimates in place of $\bar{a}_T$, $\bar{a}_N$

$$e_{\theta m}[k] = -\frac{\delta\tilde{u}_r[k]\,\delta\tilde{w}_r[k]}{C_T\,\kappa_T\,\big(\hat{\bar{a}}_T[k] - \hat{\bar{a}}_N[k]\big)}$$

$$\theta_{ym}[k] = \hat{\theta}_y[k] + e_{\theta m}[k]$$

Three axes

$$[\,\delta\tilde{u}_r \;\; \delta\tilde{v}_r \;\; \delta\tilde{w}_r\,]^{\mathsf{T}} = \hat{V}^{\mathsf{T}}\,\delta\bar{\mathbf{p}}_r$$

$$\theta_{xm}[k] = \hat{\theta}_x[k] + \frac{\delta\tilde{v}_r[k]\,\delta\tilde{w}_r[k]}{C_T\,\kappa_T\,\big(\hat{\bar{a}}_T[k] - \hat{\bar{a}}_N[k]\big)}$$

$$\theta_{ym}[k] = \hat{\theta}_y[k] - \frac{\delta\tilde{u}_r[k]\,\delta\tilde{w}_r[k]}{C_T\,\kappa_T\,\big(\hat{\bar{a}}_T[k] - \hat{\bar{a}}_N[k]\big)\cos\hat{\theta}_x[k]}$$

Part 2 Computed angle

1 Motion gain matrix in the measurement coordinate

$$\bar{A} = \bar{a}_T\, I + (\bar{a}_N - \bar{a}_T)\, \mathbf{w}\, \mathbf{w}^{\mathsf{T}}$$
$$\mathbf{w} = [\, \sin\theta_y \cos\theta_x \quad -\sin\theta_x \quad \cos\theta_y \cos\theta_x \,]^{\mathsf{T}}$$

2 Variance and covariance in the measurement coordinate

sensor noise taken equal on x, y and z

$$\begin{aligned}
\mathrm{E}\big[\delta\bar{x}_r\, \delta\bar{z}_r\big] &= C_T\, \kappa_T\, \bar{A}_{xz} \\
\mathrm{E}\big[\delta\bar{x}_r^2\big] - \mathrm{E}\big[\delta\bar{z}_r^2\big] &= C_T\, \kappa_T\, \big(\bar{A}_{xx} - \bar{A}_{zz}\big) \\
\mathrm{E}\big[\delta\bar{y}_r\, \delta\bar{z}_r\big] &= C_T\, \kappa_T\, \bar{A}_{yz} \\
\mathrm{E}\big[\delta\bar{y}_r^2\big] - \mathrm{E}\big[\delta\bar{z}_r^2\big] &= C_T\, \kappa_T\, \big(\bar{A}_{yy} - \bar{A}_{zz}\big)
\end{aligned}$$

3 Wall angle

$\theta_x = 0$

$$\begin{aligned}
\bar{A}_{xx} &= \bar{a}_T \cos^2\theta_y + \bar{a}_N \sin^2\theta_y \\
\bar{A}_{zz} &= \bar{a}_T \sin^2\theta_y + \bar{a}_N \cos^2\theta_y \\
\bar{A}_{xz} &= -(\bar{a}_T - \bar{a}_N)\, \sin\theta_y \cos\theta_y \\
\bar{A}_{xx} - \bar{A}_{zz} &= (\bar{a}_T - \bar{a}_N)\, \big(\cos^2\theta_y - \sin^2\theta_y\big) = (\bar{a}_T - \bar{a}_N)\, \cos 2\theta_y \\
-2\, \bar{A}_{xz} &= (\bar{a}_T - \bar{a}_N)\, 2 \sin\theta_y \cos\theta_y = (\bar{a}_T - \bar{a}_N)\, \sin 2\theta_y \\
\frac{-2\, \bar{A}_{xz}}{\bar{A}_{xx} - \bar{A}_{zz}} &= \frac{\sin 2\theta_y}{\cos 2\theta_y} = \tan 2\theta_y
\end{aligned}$$

θ_y is not small; no approximation

$$\begin{aligned}
\tan 2\theta_y &= \frac{-2\, \bar{A}_{xz}}{\bar{A}_{xx} - \bar{A}_{zz}} \\
\tan 2\varphi_{yz} &= \frac{-2\, \bar{A}_{yz}}{\bar{A}_{yy} - \bar{A}_{zz}} \\
\theta_x &= -\arctan\big(\tan\varphi_{yz}\, \cos\theta_y\big)
\end{aligned}$$

estimates $\hat{\bar{A}}$ in place of $\bar{A}$

$$\begin{aligned}
\tan 2\hat{\theta}_y &= \frac{-2\, \hat{\bar{A}}_{xz}}{\hat{\bar{A}}_{xx} - \hat{\bar{A}}_{zz}} \\
\tan 2\hat{\varphi}_{yz} &= \frac{-2\, \hat{\bar{A}}_{yz}}{\hat{\bar{A}}_{yy} - \hat{\bar{A}}_{zz}} \\
\hat{\theta}_x &= -\arctan\big(\tan\hat{\varphi}_{yz}\, \cos\hat{\theta}_y\big)
\end{aligned}$$

Comparison in simulation

Fig. 1 15 μm

plane wall $\theta_y = 20°$, $\theta_x = 0$, fixed point 15 μm, 20 s; left computed angle, right estimated angle; rows: $\hat{\theta}_y - \theta_y$, $\hat{\theta}_x - \theta_x$ (angle between lines, in $\pm 90°$); seed 1 dark, seeds 2–3 light

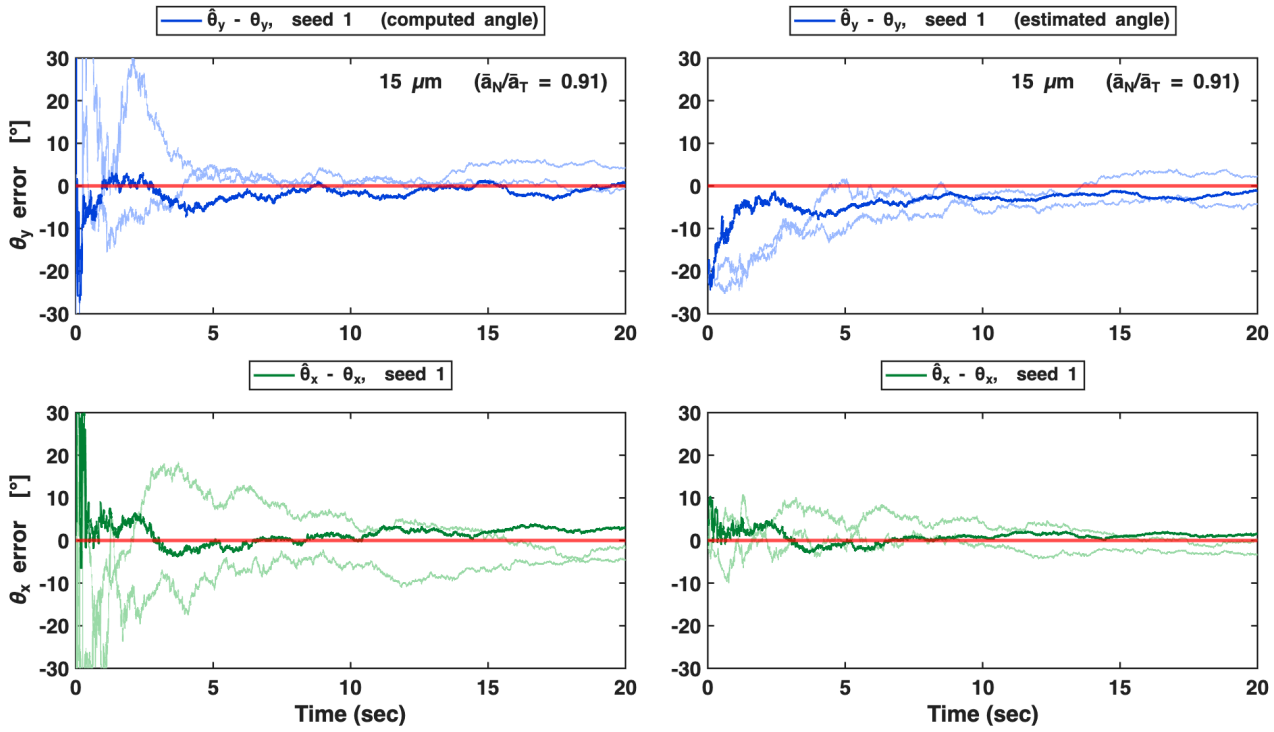


Fig. 2 50 μm

as Fig. 1 at 50 μm

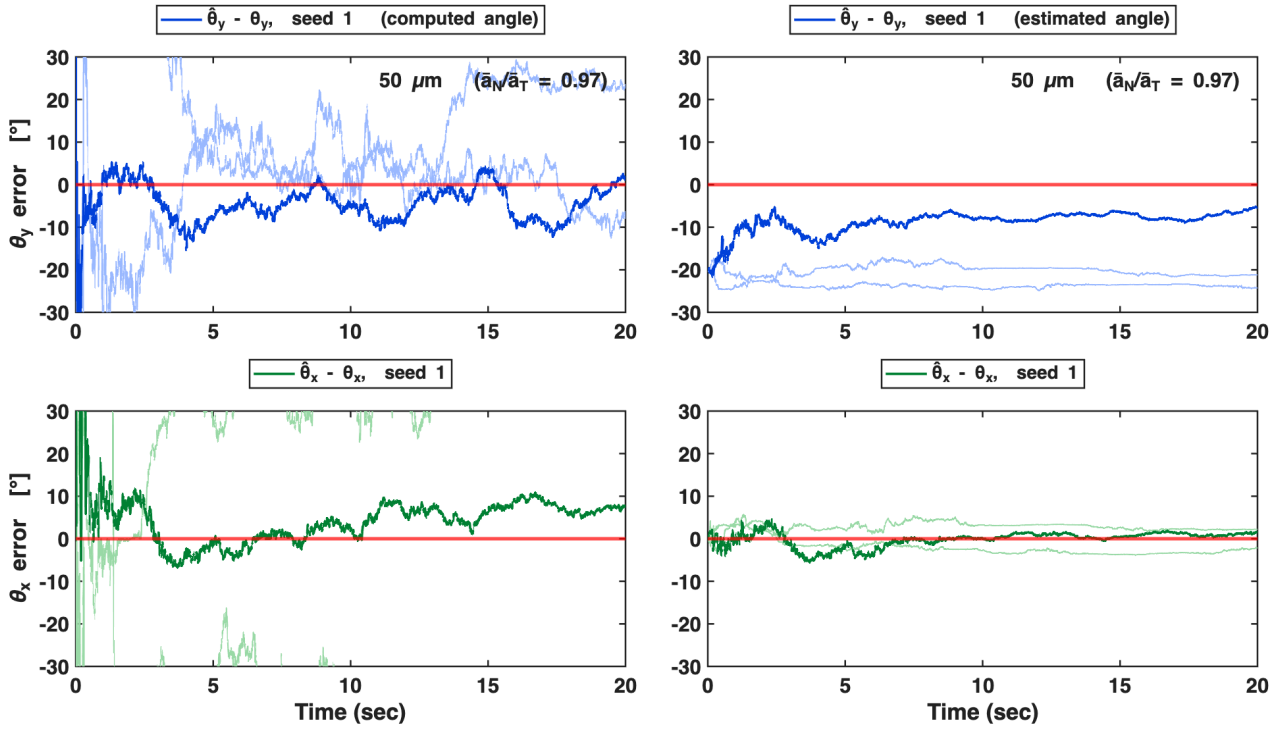


Fig. 3 Computed angle at 50 μm: numerator and denominator

left x–z plane ($\tan 2\theta_y$), right y–z plane ($\tan 2\varphi_{yz}$); rows: angle error, numerator $-2\hat{\bar{A}}_{xz}$ / $-2\hat{\bar{A}}_{yz}$, denominator $\hat{\bar{A}}_{xx} - \hat{\bar{A}}_{zz}$ / $\hat{\bar{A}}_{yy} - \hat{\bar{A}}_{zz}$ (red = true); seed 1 dark, seeds 2–3 light

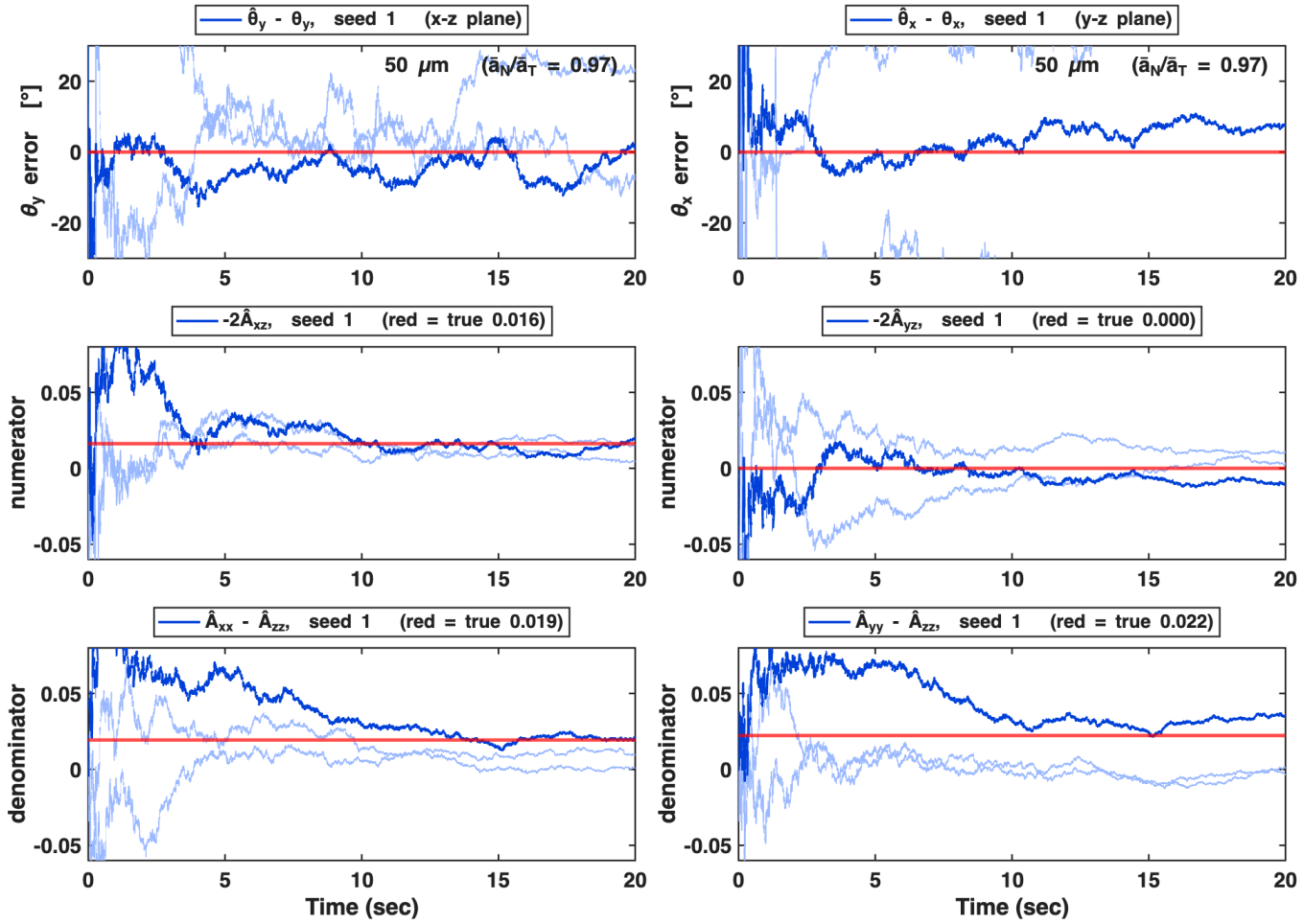


Fig. 4 Wait, regulation and scan: angles, gap and normal gain

plane wall $\theta_y = 20°$, start 15 μm, wait 5 s (no motion), regulation to $\bar{a}_N = 0.1$ ($\tau_a = 2$ s), scan along $\hat{\mathbf{u}}$ from 12 s; seed 1; left computed angle (with $\bar{A}'$), right estimated angle; rows: $\hat{\theta}_y - \theta_y$, $\hat{\theta}_x - \theta_x$, gap h (red = target), $\bar{a}_N$ (red = true)

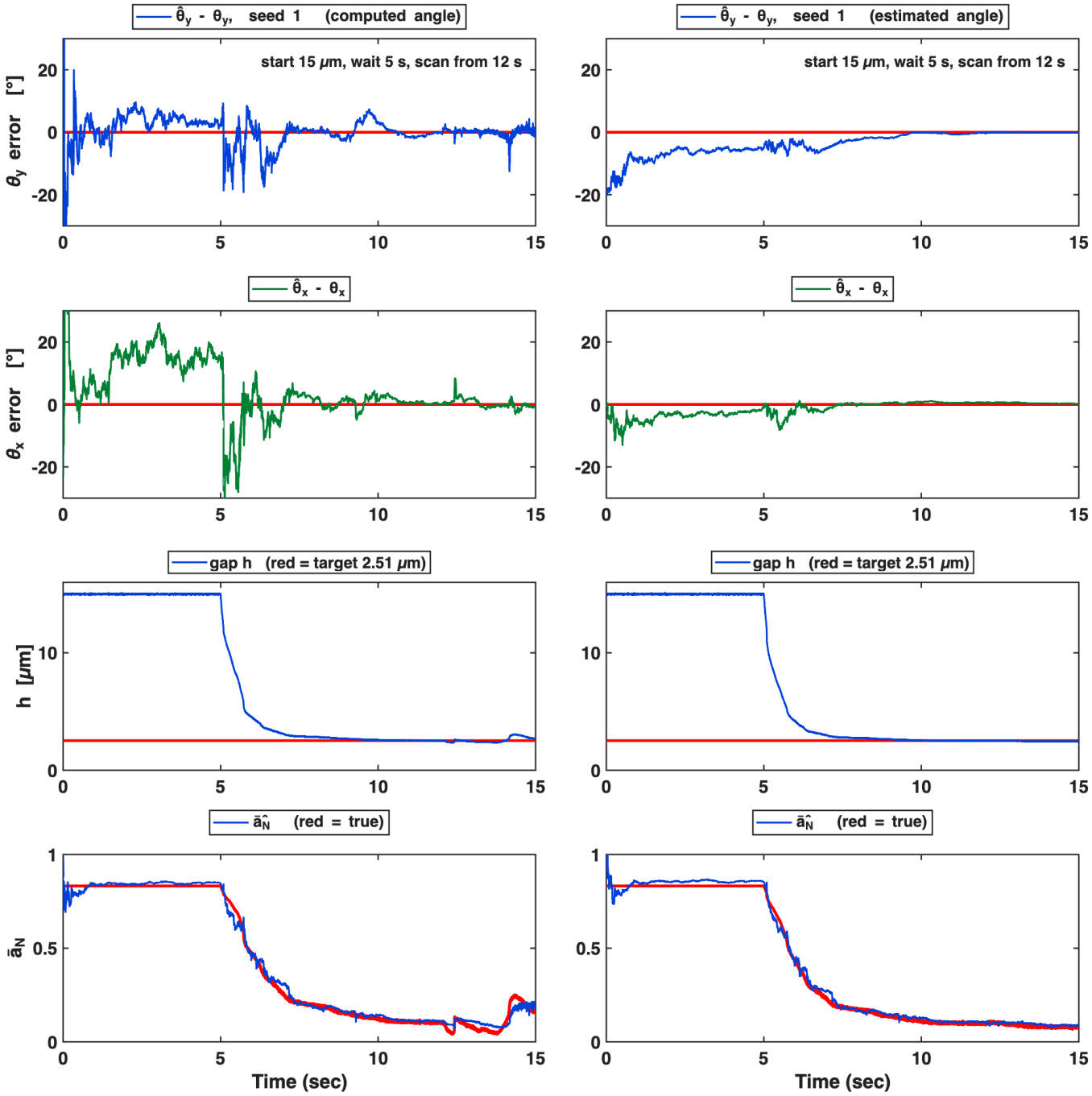


Fig. 5 Same run: wall and trajectory

top: measurement coordinate x–z, wall (gray) and probe position; bottom: local coordinate, distance along the wall u against gap h (red = target); dots: start, start of regulation, start of scan

